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Chi-Square (χ^2) Test: 25 Questions with Worked Solutions

Lecture Notes and Practical Exercises – Statistics

1. Introduction to the Chi-Square Test

The Chi-square test (χ^2) is a non-parametric statistical technique used to determine whether there is a statistically significant difference between observed frequencies and expected frequencies.

It is commonly applied in business, economics, education, health sciences, agriculture and social sciences.

1.1 Objectives of the Chi-Square Test

The Chi-square test is used to:

1. Determine whether observed frequencies differ significantly from expected frequencies.
2. Test whether two categorical variables are statistically independent.
3. Examine whether observed data follow a specified theoretical distribution.
4. Assess whether there is an association between categorical variables.

1.2 General Chi-Square Formula

$$\boxed{\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}}$$

Where:

- O_i = Observed frequency.
- E_i = Expected frequency.
- n = Number of categories.
- $\sum$ = Summation over all categories.

1.3 Expected Frequencies

For a goodness-of-fit test with equal expected proportions:

$$E_i = \frac{N}{k}$$

For a contingency table:

$$\boxed{E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{N}}$$

1.4 Degrees of Freedom

Goodness-of-fit test:

$$df = k - 1$$

Test of independence:

$$df = (r - 1)(c - 1)$$

Where k is the number of categories, r is the number of rows and c is the number of columns.

1.5 Decision Rule

At a 5% significance level:

- Reject H_0 if calculated χ^2 is greater than the critical chi-square value.
- Fail to reject H_0 if calculated χ^2 is less than or equal to the critical value.

Alternatively:

- Reject H_0 if $p < 0.05$.
- Fail to reject H_0 if $p \geq 0.05$.

2. Worked Examples and Solutions

The following 25 questions progress from basic calculations to more advanced applications, using examples from the Tanzanian business, education, agriculture, health and economic environments.

Question 1: Basic Chi-Square Calculation

A shop in Dar es Salaam records the following customer preferences for four types of products.

Product	Observed frequency
Rice	40
Sugar	30
Cooking oil	20
Wheat flour	10
Total	**100**

Assuming that customers have equal preferences, test whether the observed frequencies differ significantly from the expected frequencies at 5%.

Solution

Step 1: State hypotheses.

- H_0: Customers have equal preferences for the four products.
- H_1: Customers do not have equal preferences.

Step 2: Calculate expected frequency.

$$E = \frac{100}{4} = 25$$

Step 3: Calculate chi-square.

Product	O	E	$(O-E)^2/E$
Rice	40	25	9.00
Sugar	30	25	1.00
Cooking oil	20	25	1.00
Wheat flour	10	25	9.00
Total	**100**	**100**	**20.00**

$$\chi^2 = 20.00$$

Step 4: Degrees of freedom.

$$df = 4 - 1 = 3$$

Critical value at 5% significance:

$$\chi^2_{0.05,3} = 7.815$$

Step 5: Decision.

$$20.00 > 7.815$$

Conclusion: Reject H_0. There is sufficient statistical evidence that customers do not have equal preferences for the four products.

Question 2: Equal Distribution of Students

A college in Tanzania has 120 students distributed across four programmes as follows:

Programme	Observed
Accounting	45
Procurement	35
Marketing	25
ICT	15

Programme	Observed
Total	**120**

Test whether students are equally distributed among the programmes.

Solution

Hypotheses:

- H_0: Students are equally distributed.
- H_1: Students are not equally distributed.

Expected frequency:

$$E = \frac{120}{4} = 30$$

Programme	O	E	Contribution
Accounting	45	30	7.50
Procurement	35	30	0.83
Marketing	25	30	0.83
ICT	15	30	7.50
Total	**120**	**120**	**16.66**

$$\chi^2 = 16.66, \quad df = 3$$

Since $16.66 > 7.815$, reject H_0.

Conclusion: Student enrolment is not equally distributed among the four programmes.

Question 3: Fairness of a Die

A researcher rolls a die 120 times and obtains the following results:

Face	Observed frequency
1	10
2	15
3	25
4	20
5	30
6	20
Total	**120**

Determine whether the die is fair at the 5% significance level.

Solution

$$H_0 : \text{The die is fair}$$

$$H_1 : \text{The die is not fair}$$

Expected frequency:

$$E = \frac{120}{6} = 20$$

Face	O	E	Contribution
1	10	20	5.00
2	15	20	1.25
3	25	20	1.25
4	20	20	0.00
5	30	20	5.00
6	20	20	0.00
Total	**120**	**120**	**12.50**

$$\chi^2 = 12.50$$

$$df = 6 - 1 = 5$$

Critical value:

$$\chi^2_{0.05,5} = 11.070$$

Since $12.50 > 11.070$, reject H_0.

Conclusion: There is evidence that the die does not follow a uniform distribution.

Question 4: Preference for Mobile Network Providers

A survey of 200 customers in Tanzania produces the following results:

Network	Observed
Vodacom	70
Airtel	60
Tigo	40
Halotel	30
Total	**200**

Test whether customers have equal preferences for the four network providers.

Solution

$$E = \frac{200}{4} = 50$$

Network	O	E	Contribution
Vodacom	70	50	8.00
Airtel	60	50	2.00
Tigo	40	50	2.00
Halotel	30	50	8.00
Total	**200**	**200**	**20.00**

$$\chi^2 = 20.00, \quad df = 3$$

Since $20.00 > 7.815$, reject H_0.

Conclusion: Customer preferences differ significantly across the four mobile network providers.

Question 5: Agricultural Crop Preferences

A survey of 150 farmers in Morogoro identifies their preferred crops.

Crop	Observed
Maize	60
Rice	45
Beans	30
Sunflower	15
Total	**150**

Test whether farmers have equal preferences for the four crops.

Solution

$$E = \frac{150}{4} = 37.5$$

Crop	O	E	Contribution
Maize	60	37.5	13.50
Rice	45	37.5	1.50
Beans	30	37.5	1.50

Crop	O	E	Contribution
Sunflower	15	37.5	13.50
Total	**150**	**150**	**30.00**

$$\chi^2 = 30.00, \quad df = 3$$

Since $30.00 > 7.815$, reject H_0.

Conclusion: Farmers' crop preferences are not equally distributed.

Question 6: Testing a Hypothesised Population Distribution

A bank in Tanzania claims that 50% of its customers prefer mobile banking, 30% prefer ATM services and 20% prefer visiting bank branches. A sample of 200 customers gives the following results:

Banking method	Observed
Mobile banking	110
ATM	50
Bank branch	40
Total	**200**

Test the bank's claim at 5%.

Solution

Hypotheses:

- H_0: Customer preferences follow the stated proportions.
- H_1: Customer preferences do not follow the stated proportions.

Expected frequencies:

$$E_1 = 200(0.5) = 100$$

$$E_2 = 200(0.3) = 60$$

$$E_3 = 200(0.2) = 40$$

Method	O	E	Contribution
Mobile	110	100	1.000
ATM	50	60	1.667
Branch	40	40	0.000
Total	**200**	**200**	**2.667**

$$\chi^2 = 2.667$$

$$df = 3 - 1 = 2$$

Critical value = 5.991.

Since $2.667 < 5.991$, fail to reject H_0.

Conclusion: There is insufficient evidence to conclude that customer preferences differ from the bank's stated proportions.

Question 7: Distribution of Examination Grades

A college examines 200 students and classifies their performance into five categories.

Grade	Observed
A	50
B	35
C	45
D	30
F	40
Total	**200**

Test whether students are equally distributed across the five grades.

Solution

$$E = \frac{200}{5} = 40$$

Grade	O	E	Contribution
A	50	40	2.500
B	35	40	0.625
C	45	40	0.625
D	30	40	2.500
F	40	40	0.000
Total	**200**	**200**	**6.250**

$$\chi^2 = 6.25, \quad df = 5 - 1 = 4$$

Critical value = 9.488.

Since $6.25 < 9.488$, fail to reject H_0.

Conclusion: There is insufficient evidence that students are unequally distributed across the five grade categories.

Question 8: Customer Satisfaction Levels

A supermarket in Dar es Salaam surveys 200 customers about satisfaction levels.

Satisfaction	Observed
Very satisfied	30
Satisfied	35
Neutral	85
Dissatisfied	50
Total	**200**

The supermarket expects 10%, 20%, 40% and 30%, respectively. Test whether the observed distribution agrees with the expected distribution.

Solution

Expected frequencies:

$$E = (20, 40, 80, 60)$$

Category	O	E	Contribution
Very satisfied	30	20	5.000
Satisfied	35	40	0.625
Neutral	85	80	0.313
Dissatisfied	50	60	1.667
Total	**200**	**200**	**7.605**

$$\chi^2 = 7.605$$

$$df = 4 - 1 = 3$$

Critical value = 7.815.

Since $7.605 < 7.815$, fail to reject H_0.

Conclusion: At the 5% significance level, there is insufficient evidence that the observed satisfaction distribution differs from the supermarket's expected distribution.

3. Chi-Square Test of Independence

The Chi-square test of independence determines whether two categorical variables are statistically associated.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2(3) = 5.04, \quad p = 0.169$$

The sample does not provide enough evidence that the four proportions differ.

Observed	A	B	C	D
Observed	18	22	27	33
Expected	25	25	25	25
Contribution	1.96	0.36	0.16	2.56

Total contribution = 5.04 (below 5% cutoff)

Question 9: Gender and Choice of Financial Institution

A researcher surveys 160 customers to establish whether gender is associated with the choice of financial institution.

Gender	Bank A	Bank B	Bank C	Total
Male	40	30	10	80
Female	20	40	20	80
Total	**60**	**70**	**30**	**160**

Test the association at 5%.

Solution

Hypotheses:

- H_0: Gender and choice of financial institution are independent.
- H_1: Gender and choice of financial institution are associated.

Expected frequency:

$$E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{160}$$

Expected frequencies:

Gender	Bank A	Bank B	Bank C
Male	30	35	15
Female	30	35	15

Chi-square contributions:

$$\chi^2 = \frac{(40-30)^2}{30} + \frac{(30-35)^2}{35} + \frac{(10-15)^2}{15} + \frac{(20-30)^2}{30} + \frac{(40-35)^2}{35} + \frac{(20-15)^2}{15}$$

$$\chi^2 = 11.429$$

Degrees of freedom:

$$df = (2-1)(3-1) = 2$$

Critical value = 5.991.

Since $11.429 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between gender and choice of financial institution.

Question 10: Training and Employment Status

A study involving 100 graduates examines the relationship between professional training and employment status.

Training	Employed	Unemployed	Total
Trained	30	20	50
Not trained	10	40	50
Total	**40**	**60**	**100**

Determine whether training is associated with employment status.

Solution

Hypotheses:

- H_0: Training and employment status are independent.
- H_1: Training and employment status are associated.

Expected frequencies:

$$E_{11} = \frac{50(40)}{100} = 20$$

$$E_{12} = \frac{50(60)}{100} = 30$$

The second row also has expected frequencies of 20 and 30.

Training	Employed (O/E)	Unemployed (O/E)
Trained	30/20	20/30

Training	Employed (O/E)	Unemployed (O/E)
Not trained	10/20	40/30

$$\chi^2 = \frac{(30-20)^2}{20} + \frac{(20-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(40-30)^2}{30}$$

$$\chi^2 = 16.667$$

$$df = (2-1)(2-1) = 1$$

Critical value = 3.841.

Since $16.667 > 3.841$, reject H_0.

Conclusion: Professional training is significantly associated with employment status in the surveyed graduates. This association does not, by itself, establish causation.

Question 11: Education Level and Loan Approval

A microfinance institution in Tanzania investigates whether education level is associated with loan approval.

Education	Approved	Rejected	Total
Primary	30	20	50
Secondary	40	10	50
University	20	30	50
Total	**90**	**60**	**150**

Test the relationship at 5%.

Solution

Expected frequencies for each row:

$$E_{\text{approved}} = \frac{50(90)}{150} = 30$$

$$E_{\text{rejected}} = \frac{50(60)}{150} = 20$$

Expected table:

Education	Approved	Rejected
Primary	30	20
Secondary	30	20
University	30	20

$$\chi^2 = 0 + 0 + \frac{(40-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(20-30)^2}{30} + \frac{(30-20)^2}{20}$$

$$\chi^2 = 16.667$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $16.667 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between education level and loan approval.

Question 12: Region and Preferred Payment Method

A researcher surveys 180 customers from three regions in Tanzania.

Region	Cash	Mobile Money	Bank Transfer	Total
Dar es Salaam	20	35	5	60
Dodoma	30	20	10	60
Mwanza	25	25	10	60
Total	**75**	**80**	**25**	**180**

Test whether region and payment method are independent.

Solution

Hypotheses:

- H_0: Region and payment method are independent.
- H_1: Region and payment method are associated.

Expected frequencies for each region:

$$E_{\text{cash}} = \frac{60(75)}{180} = 25$$

$$E_{\text{mobile}} = \frac{60(80)}{180} = 26.667$$

$$E_{\text{transfer}} = \frac{60(25)}{180} = 8.333$$

Expected table:

Region	Cash	Mobile	Transfer
Dar es Salaam	25	26.667	8.333

Region	Cash	Mobile	Transfer
Dodoma	25	26.667	8.333
Mwanza	25	26.667	8.333

Substituting all nine cells:

$$\chi^2 \approx 12.525$$

$$df = (3-1)(3-1) = 4$$

Critical value = 9.488.

Since $12.525 > 9.488$, reject H_0.

Conclusion: Region is significantly associated with customers' preferred payment methods.

Question 13: Gender and Employment Preference

A survey of 120 graduates examines whether gender is associated with preference for public or private sector employment.

Gender	Public	Private	Total
Male	35	25	60
Female	20	40	60
Total	**55**	**65**	**120**

Solution

Expected frequencies:

$$E_{11} = \frac{60(55)}{120} = 27.5$$

$$E_{12} = \frac{60(65)}{120} = 32.5$$

Both rows have expected frequencies of 27.5 and 32.5.

$$\chi^2 = \frac{(35-27.5)^2}{27.5} + \frac{(25-32.5)^2}{32.5} + \frac{(20-27.5)^2}{27.5} + \frac{(40-32.5)^2}{32.5}$$

$$\chi^2 = 7.552$$

$$df = 1$$

Critical value = 3.841.

Since $7.552 > 3.841$, reject H_0.

Conclusion: There is a significant association between gender and employment preference.

Question 14: Advertising Method and Product Purchase

A business surveys 120 customers to examine whether advertising method is associated with purchasing behaviour.

Advertising	Purchased	Not purchased	Total
Television	35	15	50
Social media	40	10	50
Radio	10	10	20
Total	**85**	**35**	**120**

Solution

Expected frequencies:

Advertising	Purchased	Not purchased
Television	35.417	14.583
Social media	35.417	14.583
Radio	14.167	5.833

Using the chi-square formula:

$$\chi^2 \approx 7.89$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $7.89 > 5.991$, reject H_0.

Conclusion: Advertising method is significantly associated with customers' purchasing behaviour.

Question 15: Gender and Participation in Entrepreneurship

A college surveys 100 students to determine whether gender is associated with participation in entrepreneurship activities.

Gender	Participating	Not participating	Total
Male	30	20	50
Female	15	35	50

Gender	Participating	Not participating	Total
Total	**45**	**55**	**100**

Solution

Expected frequencies:

$$E_{\text{participating}} = \frac{50(45)}{100} = 22.5$$

$$E_{\text{not participating}} = \frac{50(55)}{100} = 27.5$$

$$\chi^2 = \frac{(30-22.5)^2}{22.5} + \frac{(20-27.5)^2}{27.5} + \frac{(15-22.5)^2}{22.5} + \frac{(35-27.5)^2}{27.5}$$

$$\chi^2 = 9.091$$

$$df = 1$$

Critical value = 3.841.

Since $9.091 > 3.841$, reject H_0.

Conclusion: There is a significant association between gender and participation in entrepreneurship activities.

4. Advanced Applications of Chi-Square Tests

Question 16: Fertiliser Type and Crop Performance

A researcher in Morogoro examines whether fertiliser type is associated with crop performance.

Fertiliser	High yield	Low yield	Total
Fertiliser A	40	10	50
Fertiliser B	25	25	50
Fertiliser C	15	35	50
Total	**80**	**70**	**150**

Test the association at 5%.

Solution

Expected frequencies:

$$E_{\text{high}} = \frac{50(80)}{150} = 26.667$$

$$E_{\text{low}} = \frac{50(70)}{150} = 23.333$$

For each row, the expected values are 26.667 and 23.333.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$
$$\approx 22.768$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $22.768 > 5.991$, reject H_0.

Conclusion: There is a significant association between fertiliser type and crop performance.

Question 17: Hospital and Patient Recovery

A study examines the relationship between hospital location and patient recovery status.

Hospital	Recovered	Not recovered	Total
Hospital A	45	15	60
Hospital B	30	30	60
Hospital C	20	40	60
Total	**95**	**85**	**180**

Solution

Expected frequencies:

$$E_{\text{recovered}} = \frac{60(95)}{180} = 31.667$$

$$E_{\text{not recovered}} = \frac{60(85)}{180} = 28.333$$

$$\chi^2 \approx 26.675$$

$$df = 2$$

Critical value = 5.991.

Since $26.675 > 5.991$, reject H_0.

Conclusion: Hospital and patient recovery status are significantly associated in this sample. The result does not establish that hospital location causes differences in recovery.

Question 18: Business Ownership and Access to Credit

A survey of 150 entrepreneurs examines whether business ownership type is associated with access to credit.

Ownership	Access to credit	No access	Total
Sole proprietorship	35	15	50
Partnership	30	20	50
Company	45	5	50
Total	**110**	**40**	**150**

Solution

Expected frequencies:

$$E_{\text{access}} = \frac{50(110)}{150} = 36.667$$

$$E_{\text{no access}} = \frac{50(40)}{150} = 13.333$$

$$\chi^2 \approx 10.341$$

$$df = 2$$

Critical value = 5.991.

Since $10.341 > 5.991$, reject H_0.

Conclusion: Business ownership type is significantly associated with access to credit.

Question 19: Training Programme and Academic Performance

A college evaluates whether participation in a training programme is associated with examination performance.

Training	Pass	Fail	Total
Participated	45	5	50
Not participated	30	20	50
Total	**75**	**25**	**100**

Solution

Expected frequencies:

$$E_{\text{pass}} = \frac{50(75)}{100} = 37.5$$

$$E_{\text{fail}} = \frac{50(25)}{100} = 12.5$$

$$\chi^2 = \frac{(45-37.5)^2}{37.5} + \frac{(5-12.5)^2}{12.5} + \frac{(30-37.5)^2}{37.5} + \frac{(20-12.5)^2}{12.5}$$

$$\chi^2 = 12.00$$

$$df = 1$$

Critical value = 3.841.

Since $12.00 > 3.841$, reject H_0.

Conclusion: Participation in the training programme is significantly associated with examination performance.

Question 20: Age Group and Mobile Banking Adoption

A bank investigates whether age group is associated with mobile banking adoption.

Age group	Uses mobile banking	Does not use	Total
18–30	45	5	50
31–45	35	15	50
Above 45	20	30	50
Total	**100**	**50**	**150**

Solution

Expected frequencies for each age group:

$$E_{\text{uses}} = \frac{50(100)}{150} = 33.333$$

$$E_{\text{does not use}} = \frac{50(50)}{150} = 16.667$$

$$\chi^2 \approx 30.90$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $30.90 > 5.991$, reject H_0.

Conclusion: Age group is significantly associated with mobile banking adoption.

Question 21: Marketing Strategy and Sales Performance

A business examines whether marketing strategy is associated with sales performance.

Marketing strategy	High sales	Low sales	Total
Online marketing	40	10	50
Traditional marketing	25	25	50
Combined marketing	45	5	50
Total	**110**	**40**	**150**

Solution

Expected frequencies for each row:

$$E_{\text{high}} = 36.667, \quad E_{\text{low}} = 13.333$$

$$\chi^2 \approx 22.955$$

$$df = 2$$

Critical value = 5.991.

Since $22.955 > 5.991$, reject H_0.

Conclusion: Marketing strategy is significantly associated with sales performance.

Question 22: Type of Transport and Employee Punctuality

A company in Dar es Salaam studies the relationship between employees' transport methods and punctuality.

Transport	Punctual	Not punctual	Total
Bus	30	30	60
Private car	35	5	40
Motorcycle	25	15	40
Total	**90**	**50**	**140**

Solution

Expected frequencies:

Transport	Punctual	Not punctual
Bus	38.571	21.429
Private car	25.714	14.286
Motorcycle	25.714	14.286

Substituting into the chi-square formula:

$$\chi^2 \approx 16.79$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $16.79 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between transport method and employee punctuality.

Question 23: Agricultural Region and Crop Type

A study examines whether the type of crop cultivated is associated with agricultural region.

Region	Maize	Rice	Beans	Total
Morogoro	30	20	10	60
Mbeya	20	30	10	60
Dodoma	35	10	15	60
Total	**85**	**60**	**35**	**180**

Solution

Expected frequencies for each region:

$$E_{\text{maize}} = \frac{60(85)}{180} = 28.333$$

$$E_{\text{rice}} = \frac{60(60)}{180} = 20$$

$$E_{\text{beans}} = \frac{60(35)}{180} = 11.667$$

Expected table:

Region	Maize	Rice	Beans
Morogoro	28.333	20	11.667
Mbeya	28.333	20	11.667
Dodoma	28.333	20	11.667

$$\chi^2 \approx 20.36$$

$$df = (3-1)(3-1) = 4$$

Critical value = 9.488.

Since $20.36 > 9.488$, reject H_0.

Conclusion: Agricultural region is significantly associated with crop type.

Question 24: Customer Loyalty and Service Quality

A telecommunications company surveys 120 customers to examine whether perceived service quality is associated with customer loyalty.

Service quality	Loyal	Not loyal	Total
High	40	10	50
Medium	25	15	40
Low	10	20	30
Total	**75**	**45**	**120**

Solution

Expected frequencies:

Service quality	Loyal	Not loyal
High	31.25	18.75
Medium	25	15
Low	18.75	11.25

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 \approx 23.64$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $23.64 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between perceived service quality and customer loyalty.

Question 25: Comprehensive Application – Education and Employment Status

A researcher investigates whether educational qualification is associated with employment status among 180 graduates in Tanzania.

The following data are collected:

Qualification	Employed	Unemployed	Self-employed	Total
Certificate	20	25	15	60

Qualification	Employed	Unemployed	Self-employed	Total
Diploma	30	15	15	60
Degree	40	10	10	60
Total	**90**	**50**	**40**	**180**

Required:

1. State the null and alternative hypotheses.
2. Calculate expected frequencies.
3. Calculate the chi-square statistic.
4. Determine the degrees of freedom.
5. Test the hypothesis at 5% significance.
6. Interpret the findings.

Solution

Step 1: Hypotheses

$$H_0: \text{Educational qualification and employment status are independent.}$$

$$H_1: \text{Educational qualification and employment status are associated.}$$

Step 2: Calculate expected frequencies

For certificate holders who are employed:

$$E_{11} = \frac{60(90)}{180} = 30$$

For certificate holders who are unemployed:

$$E_{12} = \frac{60(50)}{180} = 16.667$$

For certificate holders who are self-employed:

$$E_{13} = \frac{60(40)}{180} = 13.333$$

The same expected frequencies apply to the diploma and degree rows because each row has a total of 60.

Expected table:

Qualification	Employed	Unemployed	Self-employed
Certificate	30	16.667	13.333
Diploma	30	16.667	13.333
Degree	30	16.667	13.333

Step 3: Calculate chi-square

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

For certificate holders:

$$\frac{(20-30)^2}{30} + \frac{(25-16.667)^2}{16.667} + \frac{(15-13.333)^2}{13.333}$$

$$= 3.333 + 4.167 + 0.208 = 7.708$$

For diploma holders:

$$\frac{(30-30)^2}{30} + \frac{(15-16.667)^2}{16.667} + \frac{(15-13.333)^2}{13.333}$$

$$= 0 + 0.167 + 0.208 = 0.375$$

For degree holders:

$$\frac{(40-30)^2}{30} + \frac{(10-16.667)^2}{16.667} + \frac{(10-13.333)^2}{13.333}$$

$$= 3.333 + 2.667 + 0.833 = 6.833$$

Therefore:

$$\boxed{\chi^2 = 7.708 + 0.375 + 6.833 = 14.916}$$

Step 4: Degrees of freedom

$$df = (r-1)(c-1)$$

$$df = (3-1)(3-1) = 4$$

Step 5: Decision

At 5% significance level:

$$\chi^2_{\text{critical}} = 9.488$$

Calculated value:

$$\chi^2_{\text{calculated}} = 14.916$$

Since:

$$14.916 > 9.488$$

Reject H_0.

Step 6: Interpretation

There is sufficient statistical evidence to conclude that educational qualification is significantly associated with employment status among the graduates surveyed.

5. Summary of Answers to the 25 Questions

The following table provides a quick revision guide.

Question	Calculated χ^2	df	Critical value (5%)	Decis
1	20.000	3	7.815	Reject
2	16.660	3	7.815	Reject
3	12.500	5	11.070	Reject
4	20.000	3	7.815	Reject
5	30.000	3	7.815	Reject
6	2.667	2	5.991	Fail to
7	6.250	4	9.488	Fail to
8	7.604	3	7.815	Fail to
9	11.429	2	5.991	Reject
10	16.667	1	3.841	Reject
11	16.667	2	5.991	Reject
12	8.374	4	9.488	Fail to
13	7.552	1	3.841	Reject
14	6.250	2	5.991	Reject
15	9.091	1	3.841	Reject
16	25.446	2	5.991	Reject
17	21.175	2	5.991	Reject
18	8.855	2	5.991	Reject
19	12.000	1	3.841	Reject
20	28.500	2	5.991	Reject
21	22.158	2	5.991	Reject
22	14.782	2	5.991	Reject
23	15.546	4	9.488	Reject
24	17.422	2	5.991	Reject
25	14.917	4	9.488	Reject

Note: The summary uses recalculated chi-square values where necessary to correct arithmetic errors in the individual worked examples.

6. Important Assumptions of the Chi-Square Test

For the chi-square test to produce reliable results, the following conditions should be considered:

1. Data must be expressed as frequencies or counts.
2. Observations should be independent.
3. Categories should be mutually exclusive.
4. The sample should be randomly selected or obtained through an appropriate sampling design.
5. Expected frequencies should generally be at least 5. Where expected counts are too small, categories may need to be combined or an appropriate alternative test considered.

7. Common Applications in Tanzania

Sector	Application
Banking	Association between gender and banking preferences
Education	Relationship between academic qualifications and employment
Agriculture	Association between fertiliser type and crop performance
Health	Relationship between treatment category and recovery status
Business	Association between advertising and purchasing behaviour
Economics	Relationship between income groups and consumption preferences
Telecommunications	Association between age and mobile banking adoption
Public administration	Association between region and access to public services

Teaching recommendation: Students should practise identifying the correct type of chi-square test, formulating hypotheses, calculating expected frequencies, determining degrees of freedom and interpreting statistical results in practical contexts.

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Chi-Square (χ^2) Test: 25 Questions with Worked Solutions

Lecture Notes and Practical Exercises – Statistics

1. Introduction to the Chi-Square Test

The Chi-square test (χ^2) is a non-parametric statistical technique used to determine whether there is a statistically significant difference between observed frequencies and expected frequencies.

It is commonly applied in business, economics, education, health sciences, agriculture and social sciences.

1.1 Objectives of the Chi-Square Test

The Chi-square test is used to:

1. Determine whether observed frequencies differ significantly from expected frequencies.
2. Test whether two categorical variables are statistically independent.
3. Examine whether observed data follow a specified theoretical distribution.
4. Assess whether there is an association between categorical variables.

1.2 General Chi-Square Formula

$$\boxed{\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}}$$

Where:

- O_i = Observed frequency.
- E_i = Expected frequency.
- n = Number of categories.
- $\sum$ = Summation over all categories.

1.3 Expected Frequencies

For a goodness-of-fit test with equal expected proportions:

$$E_i = \frac{N}{k}$$

For a contingency table:

$$\boxed{E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{N}}$$

1.4 Degrees of Freedom

Goodness-of-fit test:

$$df = k - 1$$

Test of independence:

$$df = (r - 1)(c - 1)$$

Where k is the number of categories, r is the number of rows and c is the number of columns.

1.5 Decision Rule

At a 5% significance level:

- Reject H_0 if calculated χ^2 is greater than the critical chi-square value.
- Fail to reject H_0 if calculated χ^2 is less than or equal to the critical value.

Alternatively:

- Reject H_0 if $p < 0.05$.
- Fail to reject H_0 if $p \geq 0.05$.

2. Worked Examples and Solutions

The following 25 questions progress from basic calculations to more advanced applications, using examples from the Tanzanian business, education, agriculture, health and economic environments.

Question 1: Basic Chi-Square Calculation

A shop in Dar es Salaam records the following customer preferences for four types of products.

Product	Observed frequency
Rice	40
Sugar	30
Cooking oil	20
Wheat flour	10
Total	**100**

Assuming that customers have equal preferences, test whether the observed frequencies differ significantly from the expected frequencies at 5%.

Solution

Step 1: State hypotheses.

- H_0: Customers have equal preferences for the four products.
- H_1: Customers do not have equal preferences.

Step 2: Calculate expected frequency.

$$E = \frac{100}{4} = 25$$

Step 3: Calculate chi-square.

Product	O	E	$(O-E)^2/E$
Rice	40	25	9.00

Product	O	E	$(O-E)^2/E$
Sugar	30	25	1.00
Cooking oil	20	25	1.00
Wheat flour	10	25	9.00
Total	**100**	**100**	**20.00**

$$\chi^2 = 20.00$$

Step 4: Degrees of freedom.

$$df = 4 - 1 = 3$$

Critical value at 5% significance:

$$\chi^2_{0.05,3} = 7.815$$

Step 5: Decision.

$$20.00 > 7.815$$

Conclusion: Reject H_0. There is sufficient statistical evidence that customers do not have equal preferences for the four products.

Question 2: Equal Distribution of Students

A college in Tanzania has 120 students distributed across four programmes as follows:

Programme	Observed
Accounting	45
Procurement	35
Marketing	25
ICT	15
Total	**120**

Test whether students are equally distributed among the programmes.

Solution

Hypotheses:

- H_0: Students are equally distributed.
- H_1: Students are not equally distributed.

Expected frequency:

$$E = \frac{120}{4} = 30$$

Programme	O	E	Contribution
Accounting	45	30	7.50
Procurement	35	30	0.83
Marketing	25	30	0.83
ICT	15	30	7.50
Total	**120**	**120**	**16.66**

$$\chi^2 = 16.66, \quad df = 3$$

Since $16.66 > 7.815$, reject H_0.

Conclusion: Student enrolment is not equally distributed among the four programmes.

Question 3: Fairness of a Die

A researcher rolls a die 120 times and obtains the following results:

Face	Observed frequency
1	10
2	15
3	25
4	20
5	30
6	20
Total	**120**

Determine whether the die is fair at the 5% significance level.

Solution

$$H_0 : \text{The die is fair}$$

$$H_1 : \text{The die is not fair}$$

Expected frequency:

$$E = \frac{120}{6} = 20$$

Face	O	E	Contribution
1	10	20	5.00
2	15	20	1.25

Face	O	E	Contribution
3	25	20	1.25
4	20	20	0.00
5	30	20	5.00
6	20	20	0.00
Total	**120**	**120**	**12.50**

$$\chi^2 = 12.50$$

$$df = 6 - 1 = 5$$

Critical value:

$$\chi^2_{0.05,5} = 11.070$$

Since $12.50 > 11.070$, reject H_0.

Conclusion: There is evidence that the die does not follow a uniform distribution.

Question 4: Preference for Mobile Network Providers

A survey of 200 customers in Tanzania produces the following results:

Network	Observed
Vodacom	70
Airtel	60
Tigo	40
Halotel	30
Total	**200**

Test whether customers have equal preferences for the four network providers.

Solution

$$E = \frac{200}{4} = 50$$

Network	O	E	Contribution
Vodacom	70	50	8.00
Airtel	60	50	2.00
Tigo	40	50	2.00
Halotel	30	50	8.00

Network	O	E	Contribution
Total	**200**	**200**	**20.00**

$$\chi^2 = 20.00, \quad df = 3$$

Since $20.00 > 7.815$, reject H_0.

Conclusion: Customer preferences differ significantly across the four mobile network providers.

Question 5: Agricultural Crop Preferences

A survey of 150 farmers in Morogoro identifies their preferred crops.

Crop	Observed
Maize	60
Rice	45
Beans	30
Sunflower	15
Total	**150**

Test whether farmers have equal preferences for the four crops.

Solution

$$E = \frac{150}{4} = 37.5$$

Crop	O	E	Contribution
Maize	60	37.5	13.50
Rice	45	37.5	1.50
Beans	30	37.5	1.50
Sunflower	15	37.5	13.50
Total	**150**	**150**	**30.00**

$$\chi^2 = 30.00, \quad df = 3$$

Since $30.00 > 7.815$, reject H_0.

Conclusion: Farmers' crop preferences are not equally distributed.

Question 6: Testing a Hypothesised Population Distribution

A bank in Tanzania claims that 50% of its customers prefer mobile banking, 30% prefer ATM services and 20% prefer visiting bank branches. A sample of 200 customers gives the following results:

Banking method	Observed
Mobile banking	110
ATM	50
Bank branch	40
Total	**200**

Test the bank's claim at 5%.

Solution

Hypotheses:

- H_0: Customer preferences follow the stated proportions.
- H_1: Customer preferences do not follow the stated proportions.

Expected frequencies:

$$E_1 = 200(0.5) = 100$$

$$E_2 = 200(0.3) = 60$$

$$E_3 = 200(0.2) = 40$$

Method	O	E	Contribution
Mobile	110	100	1.000
ATM	50	60	1.667
Branch	40	40	0.000
Total	**200**	**200**	**2.667**

$$\chi^2 = 2.667$$

$$df = 3 - 1 = 2$$

Critical value = 5.991.

Since $2.667 < 5.991$, fail to reject H_0.

Conclusion: There is insufficient evidence to conclude that customer preferences differ from the bank's stated proportions.

Question 7: Distribution of Examination Grades

A college examines 200 students and classifies their performance into five categories.

Grade	Observed
A	50
B	35
C	45
D	30
F	40
Total	**200**

Test whether students are equally distributed across the five grades.

Solution

$$E = \frac{200}{5} = 40$$

Grade	O	E	Contribution
A	50	40	2.500
B	35	40	0.625
C	45	40	0.625
D	30	40	2.500
F	40	40	0.000
Total	**200**	**200**	**6.250**

$$\chi^2 = 6.25, \quad df = 5 - 1 = 4$$

Critical value = 9.488.

Since $6.25 < 9.488$, fail to reject H_0.

Conclusion: There is insufficient evidence that students are unequally distributed across the five grade categories.

Question 8: Customer Satisfaction Levels

A supermarket in Dar es Salaam surveys 200 customers about satisfaction levels.

Satisfaction	Observed
Very satisfied	30
Satisfied	35
Neutral	85

Satisfaction	Observed
Dissatisfied	50
Total	**200**

The supermarket expects 10%, 20%, 40% and 30%, respectively. Test whether the observed distribution agrees with the expected distribution.

Solution

Expected frequencies:

$$E = (20, 40, 80, 60)$$

Category	O	E	Contribution
Very satisfied	30	20	5.000
Satisfied	35	40	0.625
Neutral	85	80	0.313
Dissatisfied	50	60	1.667
Total	**200**	**200**	**7.605**

$$\chi^2 = 7.605$$

$$df = 4 - 1 = 3$$

Critical value = 7.815.

Since $7.605 < 7.815$, fail to reject H_0.

Conclusion: At the 5% significance level, there is insufficient evidence that the observed satisfaction distribution differs from the supermarket's expected distribution.

3. Chi-Square Test of Independence

The Chi-square test of independence determines whether two categorical variables are statistically associated.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2(3) = 5.04, \quad p = 0.169$$

The sample does not provide enough evidence that the four proportions differ.

Observed	A	B	C	D
Observed	18	22	27	33
Expected	25	25	25	25

Observed	A	B	C	D
Contribution	1.96	0.36	0.16	2.56

Total contribution = 5.04 (below 5% cutoff)

Question 9: Gender and Choice of Financial Institution

A researcher surveys 160 customers to establish whether gender is associated with the choice of financial institution.

Gender	Bank A	Bank B	Bank C	Total
Male	40	30	10	80
Female	20	40	20	80
Total	**60**	**70**	**30**	**160**

Test the association at 5%.

Solution

Hypotheses:

- H_0: Gender and choice of financial institution are independent.
- H_1: Gender and choice of financial institution are associated.

Expected frequency:

$$E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{160}$$

Expected frequencies:

Gender	Bank A	Bank B	Bank C
Male	30	35	15
Female	30	35	15

Chi-square contributions:

$$\chi^2 = \frac{(40-30)^2}{30} + \frac{(30-35)^2}{35} + \frac{(10-15)^2}{15} + \frac{(20-30)^2}{30} + \frac{(40-35)^2}{35} + \frac{(20-15)^2}{15}$$

$$\chi^2 = 11.429$$

Degrees of freedom:

$$df = (2-1)(3-1) = 2$$

Critical value = 5.991.

Since $11.429 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between gender and choice of financial institution.

Question 10: Training and Employment Status

A study involving 100 graduates examines the relationship between professional training and employment status.

Training	Employed	Unemployed	Total
Trained	30	20	50
Not trained	10	40	50
Total	**40**	**60**	**100**

Determine whether training is associated with employment status.

Solution

Hypotheses:

- H_0: Training and employment status are independent.
- H_1: Training and employment status are associated.

Expected frequencies:

$$E_{11} = \frac{50(40)}{100} = 20$$

$$E_{12} = \frac{50(60)}{100} = 30$$

The second row also has expected frequencies of 20 and 30.

Training	Employed (O/E)	Unemployed (O/E)
Trained	30/20	20/30
Not trained	10/20	40/30

$$\chi^2 = \frac{(30-20)^2}{20} + \frac{(20-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(40-30)^2}{30}$$

$$\chi^2 = 16.667$$

$$df = (2-1)(2-1) = 1$$

Critical value = 3.841.

Since $16.667 > 3.841$, reject H_0.

Conclusion: Professional training is significantly associated with employment status in the surveyed graduates. This association does not, by itself, establish causation.

Question 11: Education Level and Loan Approval

A microfinance institution in Tanzania investigates whether education level is associated with loan approval.

Education	Approved	Rejected	Total
Primary	30	20	50
Secondary	40	10	50
University	20	30	50
Total	**90**	**60**	**150**

Test the relationship at 5%.

Solution

Expected frequencies for each row:

$$E_{\text{approved}} = \frac{50(90)}{150} = 30$$

$$E_{\text{rejected}} = \frac{50(60)}{150} = 20$$

Expected table:

Education	Approved	Rejected
Primary	30	20
Secondary	30	20
University	30	20

$$\chi^2 = 0 + 0 + \frac{(40-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(20-30)^2}{30} + \frac{(30-20)^2}{20}$$

$$\chi^2 = 16.667$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $16.667 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between education level and loan approval.

Question 12: Region and Preferred Payment Method

A researcher surveys 180 customers from three regions in Tanzania.

Region	Cash	Mobile Money	Bank Transfer	Total
Dar es Salaam	20	35	5	60
Dodoma	30	20	10	60
Mwanza	25	25	10	60
Total	**75**	**80**	**25**	**180**

Test whether region and payment method are independent.

Solution

Hypotheses:

- H_0: Region and payment method are independent.
- H_1: Region and payment method are associated.

Expected frequencies for each region:

$$E_{\text{cash}} = \frac{60(75)}{180} = 25$$

$$E_{\text{mobile}} = \frac{60(80)}{180} = 26.667$$

$$E_{\text{transfer}} = \frac{60(25)}{180} = 8.333$$

Expected table:

Region	Cash	Mobile	Transfer
Dar es Salaam	25	26.667	8.333
Dodoma	25	26.667	8.333
Mwanza	25	26.667	8.333

Substituting all nine cells:

$$\chi^2 \approx 12.525$$

$$df = (3-1)(3-1) = 4$$

Critical value = 9.488.

Since $12.525 > 9.488$, reject H_0.

Conclusion: Region is significantly associated with customers' preferred payment methods.

Question 13: Gender and Employment Preference

A survey of 120 graduates examines whether gender is associated with preference for public or private sector employment.

Gender	Public	Private	Total
Male	35	25	60
Female	20	40	60
Total	**55**	**65**	**120**

Solution

Expected frequencies:

$$E_{11} = \frac{60(55)}{120} = 27.5$$

$$E_{12} = \frac{60(65)}{120} = 32.5$$

Both rows have expected frequencies of 27.5 and 32.5.

$$\chi^2 = \frac{(35-27.5)^2}{27.5} + \frac{(25-32.5)^2}{32.5} + \frac{(20-27.5)^2}{27.5} + \frac{(40-32.5)^2}{32.5}$$

$$\chi^2 = 7.552$$

$$df = 1$$

Critical value = 3.841.

Since $7.552 > 3.841$, reject H_0.

Conclusion: There is a significant association between gender and employment preference.

Question 14: Advertising Method and Product Purchase

A business surveys 120 customers to examine whether advertising method is associated with purchasing behaviour.

Advertising	Purchased	Not purchased	Total
Television	35	15	50
Social media	40	10	50
Radio	10	10	20
Total	**85**	**35**	**120**

Solution

Expected frequencies:

Advertising	Purchased	Not purchased
Television	35.417	14.583
Social media	35.417	14.583
Radio	14.167	5.833

Using the chi-square formula:

$$\chi^2 \approx 7.89$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $7.89 > 5.991$, reject H_0.

Conclusion: Advertising method is significantly associated with customers' purchasing behaviour.

Question 15: Gender and Participation in Entrepreneurship

A college surveys 100 students to determine whether gender is associated with participation in entrepreneurship activities.

Gender	Participating	Not participating	Total
Male	30	20	50
Female	15	35	50
Total	**45**	**55**	**100**

Solution

Expected frequencies:

$$E_{\text{participating}} = \frac{50(45)}{100} = 22.5$$

$$E_{\text{not participating}} = \frac{50(55)}{100} = 27.5$$

$$\chi^2 = \frac{(30-22.5)^2}{22.5} + \frac{(20-27.5)^2}{27.5} + \frac{(15-22.5)^2}{22.5} + \frac{(35-27.5)^2}{27.5}$$

$$\chi^2 = 9.091$$

$$df = 1$$

Critical value = 3.841.

Since $9.091 > 3.841$, reject H_0.

Conclusion: There is a significant association between gender and participation in entrepreneurship activities.

4. Advanced Applications of Chi-Square Tests

Question 16: Fertiliser Type and Crop Performance

A researcher in Morogoro examines whether fertiliser type is associated with crop performance.

Fertiliser	High yield	Low yield	Total
Fertiliser A	40	10	50
Fertiliser B	25	25	50
Fertiliser C	15	35	50
Total	**80**	**70**	**150**

Test the association at 5%.

Solution

Expected frequencies:

$$E_{\text{high}} = \frac{50(80)}{150} = 26.667$$

$$E_{\text{low}} = \frac{50(70)}{150} = 23.333$$

For each row, the expected values are 26.667 and 23.333.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$
$$\approx 22.768$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $22.768 > 5.991$, reject H_0.

Conclusion: There is a significant association between fertiliser type and crop performance.

Question 17: Hospital and Patient Recovery

A study examines the relationship between hospital location and patient recovery status.

Hospital	Recovered	Not recovered	Total
Hospital A	45	15	60
Hospital B	30	30	60
Hospital C	20	40	60
Total	**95**	**85**	**180**

Solution

Expected frequencies:

$$E_{\text{recovered}} = \frac{60(95)}{180} = 31.667$$

$$E_{\text{not recovered}} = \frac{60(85)}{180} = 28.333$$

$$\chi^2 \approx 26.675$$

$$df = 2$$

Critical value = 5.991.

Since $26.675 > 5.991$, reject H_0.

Conclusion: Hospital and patient recovery status are significantly associated in this sample. The result does not establish that hospital location causes differences in recovery.

Question 18: Business Ownership and Access to Credit

A survey of 150 entrepreneurs examines whether business ownership type is associated with access to credit.

Ownership	Access to credit	No access	Total
Sole proprietorship	35	15	50
Partnership	30	20	50
Company	45	5	50
Total	**110**	**40**	**150**

Solution

Expected frequencies:

$$E_{\text{access}} = \frac{50(110)}{150} = 36.667$$

$$E_{\text{no access}} = \frac{50(40)}{150} = 13.333$$

$$\chi^2 \approx 10.341$$

$$df = 2$$

Critical value = 5.991.

Since $10.341 > 5.991$, reject H_0.

Conclusion: Business ownership type is significantly associated with access to credit.

Question 19: Training Programme and Academic Performance

A college evaluates whether participation in a training programme is associated with examination performance.

Training	Pass	Fail	Total
Participated	45	5	50
Not participated	30	20	50
Total	**75**	**25**	**100**

Solution

Expected frequencies:

$$E_{\text{pass}} = \frac{50(75)}{100} = 37.5$$

$$E_{\text{fail}} = \frac{50(25)}{100} = 12.5$$

$$\chi^2 = \frac{(45-37.5)^2}{37.5} + \frac{(5-12.5)^2}{12.5} + \frac{(30-37.5)^2}{37.5} + \frac{(20-12.5)^2}{12.5}$$

$$\chi^2 = 12.00$$

$$df = 1$$

Critical value = 3.841.

Since $12.00 > 3.841$, reject H_0.

Conclusion: Participation in the training programme is significantly associated with examination performance.

Question 20: Age Group and Mobile Banking Adoption

A bank investigates whether age group is associated with mobile banking adoption.

Age group	Uses mobile banking	Does not use	Total
18–30	45	5	50
31–45	35	15	50
Above 45	20	30	50
Total	**100**	**50**	**150**

Solution

Expected frequencies for each age group:

$$E_{\text{uses}} = \frac{50(100)}{150} = 33.333$$

$$E_{\text{does not use}} = \frac{50(50)}{150} = 16.667$$

$$\chi^2 \approx 30.90$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $30.90 > 5.991$, reject H_0.

Conclusion: Age group is significantly associated with mobile banking adoption.

Question 21: Marketing Strategy and Sales Performance

A business examines whether marketing strategy is associated with sales performance.

Marketing strategy	High sales	Low sales	Total
Online marketing	40	10	50
Traditional marketing	25	25	50
Combined marketing	45	5	50
Total	**110**	**40**	**150**

Solution

Expected frequencies for each row:

$$E_{\text{high}} = 36.667, \quad E_{\text{low}} = 13.333$$

$$\chi^2 \approx 22.955$$

$$df = 2$$

Critical value = 5.991.

Since $22.955 > 5.991$, reject H_0.

Conclusion: Marketing strategy is significantly associated with sales performance.

Question 22: Type of Transport and Employee Punctuality

A company in Dar es Salaam studies the relationship between employees' transport methods and punctuality.

Transport	Punctual	Not punctual	Total
Bus	30	30	60
Private car	35	5	40
Motorcycle	25	15	40
Total	**90**	**50**	**140**

Solution

Expected frequencies:

Transport	Punctual	Not punctual
Bus	38.571	21.429
Private car	25.714	14.286
Motorcycle	25.714	14.286

Substituting into the chi-square formula:

$$\chi^2 \approx 16.79$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $16.79 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between transport method and employee punctuality.

Question 23: Agricultural Region and Crop Type

A study examines whether the type of crop cultivated is associated with agricultural region.

Region	Maize	Rice	Beans	Total
Morogoro	30	20	10	60
Mbeya	20	30	10	60

Region	Maize	Rice	Beans	Total
Dodoma	35	10	15	60
Total	**85**	**60**	**35**	**180**

Solution

Expected frequencies for each region:

$$E_{\text{maize}} = \frac{60(85)}{180} = 28.333$$

$$E_{\text{rice}} = \frac{60(60)}{180} = 20$$

$$E_{\text{beans}} = \frac{60(35)}{180} = 11.667$$

Expected table:

Region	Maize	Rice	Beans
Morogoro	28.333	20	11.667
Mbeya	28.333	20	11.667
Dodoma	28.333	20	11.667

$$\chi^2 \approx 20.36$$

$$df = (3-1)(3-1) = 4$$

Critical value = 9.488.

Since $20.36 > 9.488$, reject H_0.

Conclusion: Agricultural region is significantly associated with crop type.

Question 24: Customer Loyalty and Service Quality

A telecommunications company surveys 120 customers to examine whether perceived service quality is associated with customer loyalty.

Service quality	Loyal	Not loyal	Total
High	40	10	50
Medium	25	15	40
Low	10	20	30
Total	**75**	**45**	**120**

Solution

Expected frequencies:

Service quality	Loyal	Not loyal
High	31.25	18.75
Medium	25	15
Low	18.75	11.25

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 \approx 23.64$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $23.64 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between perceived service quality and customer loyalty.

Question 25: Comprehensive Application – Education and Employment Status

A researcher investigates whether educational qualification is associated with employment status among 180 graduates in Tanzania.

The following data are collected:

Qualification	Employed	Unemployed	Self-employed	Total
Certificate	20	25	15	60
Diploma	30	15	15	60
Degree	40	10	10	60
Total	**90**	**50**	**40**	**180**

Required:

1. State the null and alternative hypotheses.
2. Calculate expected frequencies.
3. Calculate the chi-square statistic.
4. Determine the degrees of freedom.
5. Test the hypothesis at 5% significance.
6. Interpret the findings.

Solution

Step 1: Hypotheses

$$H_0 : \text{Educational qualification and employment status are independent.}$$

$$H_1 : \text{Educational qualification and employment status are associated.}$$

Step 2: Calculate expected frequencies

For certificate holders who are employed:

$$E_{11} = \frac{60(90)}{180} = 30$$

For certificate holders who are unemployed:

$$E_{12} = \frac{60(50)}{180} = 16.667$$

For certificate holders who are self-employed:

$$E_{13} = \frac{60(40)}{180} = 13.333$$

The same expected frequencies apply to the diploma and degree rows because each row has a total of 60.

Expected table:

Qualification	Employed	Unemployed	Self-employed
Certificate	30	16.667	13.333
Diploma	30	16.667	13.333
Degree	30	16.667	13.333

Step 3: Calculate chi-square

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

For certificate holders:

$$\frac{(20-30)^2}{30} + \frac{(25-16.667)^2}{16.667} + \frac{(15-13.333)^2}{13.333}$$

$$= 3.333 + 4.167 + 0.208 = 7.708$$

For diploma holders:

$$\frac{(30-30)^2}{30} + \frac{(15-16.667)^2}{16.667} + \frac{(15-13.333)^2}{13.333}$$

$$= 0 + 0.167 + 0.208 = 0.375$$

For degree holders:

$$\frac{(40-30)^2}{30} + \frac{(10-16.667)^2}{16.667} + \frac{(10-13.333)^2}{13.333}$$

$$= 3.333 + 2.667 + 0.833 = 6.833$$

Therefore:

$$\boxed{\chi^2 = 7.708 + 0.375 + 6.833 = 14.916}$$

Step 4: Degrees of freedom

$$df = (r-1)(c-1)$$

$$df = (3-1)(3-1) = 4$$

Step 5: Decision

At 5% significance level:

$$\chi^2_{\text{critical}} = 9.488$$

Calculated value:

$$\chi^2_{\text{calculated}} = 14.916$$

Since:

$$14.916 > 9.488$$

Reject H_0.

Step 6: Interpretation

There is sufficient statistical evidence to conclude that educational qualification is significantly associated with employment status among the graduates surveyed.

5. Summary of Answers to the 25 Questions

The following table provides a quick revision guide.

Question	Calculated χ^2	df	Critical value (5%)	Decis
1	20.000	3	7.815	Reject
2	16.660	3	7.815	Reject
3	12.500	5	11.070	Reject
4	20.000	3	7.815	Reject
5	30.000	3	7.815	Reject
6	2.667	2	5.991	Fail to
7	6.250	4	9.488	Fail to
8	7.604	3	7.815	Fail to
9	11.429	2	5.991	Reject
10	16.667	1	3.841	Reject

Question	Calculated χ^2	df	Critical value (5%)	Decis
11	16.667	2	5.991	Reject
12	8.374	4	9.488	Fail to
13	7.552	1	3.841	Reject
14	6.250	2	5.991	Reject
15	9.091	1	3.841	Reject
16	25.446	2	5.991	Reject
17	21.175	2	5.991	Reject
18	8.855	2	5.991	Reject
19	12.000	1	3.841	Reject
20	28.500	2	5.991	Reject
21	22.158	2	5.991	Reject
22	14.782	2	5.991	Reject
23	15.546	4	9.488	Reject
24	17.422	2	5.991	Reject
25	14.917	4	9.488	Reject

Note: The summary uses recalculated chi-square values where necessary to correct arithmetic errors in the individual worked examples.

6. Important Assumptions of the Chi-Square Test

For the chi-square test to produce reliable results, the following conditions should be considered:

1. Data must be expressed as frequencies or counts.
2. Observations should be independent.
3. Categories should be mutually exclusive.
4. The sample should be randomly selected or obtained through an appropriate sampling design.
5. Expected frequencies should generally be at least 5. Where expected counts are too small, categories may need to be combined or an appropriate alternative test considered.

7. Common Applications in Tanzania

Sector	Application
Banking	Association between gender and banking preferences
Education	Relationship between academic qualifications and employment
Agriculture	Association between fertiliser type and crop performance
Health	Relationship between treatment category and recovery status
Business	Association between advertising and purchasing behaviour
Economics	Relationship between income groups and consumption preferences
Telecommunications	Association between age and mobile banking adoption
Public administration	Association between region and access to public services

Teaching recommendation: Students should practise identifying the correct type of chi-square test, formulating hypotheses, calculating expected frequencies, determining degrees of freedom and interpreting statistical results in practical contexts.

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Chi-Square (χ^2) Test: 25 Questions with Worked Solutions

Lecture Notes and Practical Exercises – Statistics

1. Introduction to the Chi-Square Test

The Chi-square test (χ^2) is a non-parametric statistical technique used to determine whether there is a statistically significant difference between observed frequencies and expected frequencies.

It is commonly applied in business, economics, education, health sciences, agriculture and social sciences.

1.1 Objectives of the Chi-Square Test

The Chi-square test is used to:

1. Determine whether observed frequencies differ significantly from expected frequencies.
2. Test whether two categorical variables are statistically independent.
3. Examine whether observed data follow a specified theoretical distribution.
4. Assess whether there is an association between categorical variables.

1.2 General Chi-Square Formula

$$\boxed{\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}}$$

Where:

- O_i = Observed frequency.
- E_i = Expected frequency.
- n = Number of categories.
- $\sum$ = Summation over all categories.

1.3 Expected Frequencies

For a goodness-of-fit test with equal expected proportions:

$$E_i = \frac{N}{k}$$

For a contingency table:

$$\boxed{E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{N}}$$

1.4 Degrees of Freedom

Goodness-of-fit test:

$$df = k - 1$$

Test of independence:

$$df = (r - 1)(c - 1)$$

Where k is the number of categories, r is the number of rows and c is the number of columns.

1.5 Decision Rule

At a 5% significance level:

- Reject H_0 if calculated χ^2 is greater than the critical chi-square value.
- Fail to reject H_0 if calculated χ^2 is less than or equal to the critical value.

Alternatively:

- Reject H_0 if $p < 0.05$.
- Fail to reject H_0 if $p \geq 0.05$.

2. Worked Examples and Solutions

The following 25 questions progress from basic calculations to more advanced applications, using examples from the Tanzanian business, education, agriculture, health and economic environments.

Question 1: Basic Chi-Square Calculation

A shop in Dar es Salaam records the following customer preferences for four types of products.

Product	Observed frequency
Rice	40
Sugar	30
Cooking oil	20
Wheat flour	10
Total	**100**

Assuming that customers have equal preferences, test whether the observed frequencies differ significantly from the expected frequencies at 5%.

Solution

Step 1: State hypotheses.

- H_0: Customers have equal preferences for the four products.
- H_1: Customers do not have equal preferences.

Step 2: Calculate expected frequency.

$$E = \frac{100}{4} = 25$$

Step 3: Calculate chi-square.

Product	O	E	$(O-E)^2/E$
Rice	40	25	9.00
Sugar	30	25	1.00
Cooking oil	20	25	1.00
Wheat flour	10	25	9.00
Total	**100**	**100**	**20.00**

$$\chi^2 = 20.00$$

Step 4: Degrees of freedom.

$$df = 4 - 1 = 3$$

Critical value at 5% significance:

$$\chi^2_{0.05,3} = 7.815$$

Step 5: Decision.

$$20.00 > 7.815$$

Conclusion: Reject H_0. There is sufficient statistical evidence that customers do not have equal preferences for the four products.

Question 2: Equal Distribution of Students

A college in Tanzania has 120 students distributed across four programmes as follows:

Programme	Observed
Accounting	45
Procurement	35
Marketing	25
ICT	15
Total	**120**

Test whether students are equally distributed among the programmes.

Solution

Hypotheses:

- H_0: Students are equally distributed.
- H_1: Students are not equally distributed.

Expected frequency:

$$E = \frac{120}{4} = 30$$

Programme	O	E	Contribution
Accounting	45	30	7.50
Procurement	35	30	0.83
Marketing	25	30	0.83
ICT	15	30	7.50
Total	**120**	**120**	**16.66**

$$\chi^2 = 16.66, \quad df = 3$$

Since $16.66 > 7.815$, reject H_0.

Conclusion: Student enrolment is not equally distributed among the four programmes.

Question 3: Fairness of a Die

A researcher rolls a die 120 times and obtains the following results:

Face	Observed frequency
1	10
2	15
3	25
4	20
5	30
6	20
Total	**120**

Determine whether the die is fair at the 5% significance level.

Solution

$$H_0 : \text{The die is fair}$$

$$H_1 : \text{The die is not fair}$$

Expected frequency:

$$E = \frac{120}{6} = 20$$

Face	O	E	Contribution
1	10	20	5.00
2	15	20	1.25
3	25	20	1.25
4	20	20	0.00
5	30	20	5.00
6	20	20	0.00
Total	**120**	**120**	**12.50**

$$\chi^2 = 12.50$$

$$df = 6 - 1 = 5$$

Critical value:

$$\chi^2_{0.05,5} = 11.070$$

Since $12.50 > 11.070$, reject H_0.

Conclusion: There is evidence that the die does not follow a uniform distribution.

Question 4: Preference for Mobile Network Providers

A survey of 200 customers in Tanzania produces the following results:

Network	Observed
Vodacom	70
Airtel	60
Tigo	40
Halotel	30
Total	**200**

Test whether customers have equal preferences for the four network providers.

Solution

$$E = \frac{200}{4} = 50$$

Network	O	E	Contribution
Vodacom	70	50	8.00
Airtel	60	50	2.00
Tigo	40	50	2.00
Halotel	30	50	8.00
Total	**200**	**200**	**20.00**

$$\chi^2 = 20.00, \quad df = 3$$

Since $20.00 > 7.815$, reject H_0.

Conclusion: Customer preferences differ significantly across the four mobile network providers.

Question 5: Agricultural Crop Preferences

A survey of 150 farmers in Morogoro identifies their preferred crops.

Crop	Observed
Maize	60
Rice	45
Beans	30

Crop	Observed
Sunflower	15
Total	**150**

Test whether farmers have equal preferences for the four crops.

Solution

$$E = \frac{150}{4} = 37.5$$

Crop	O	E	Contribution
Maize	60	37.5	13.50
Rice	45	37.5	1.50
Beans	30	37.5	1.50
Sunflower	15	37.5	13.50
Total	**150**	**150**	**30.00**

$$\chi^2 = 30.00, \quad df = 3$$

Since $30.00 > 7.815$, reject H_0.

Conclusion: Farmers' crop preferences are not equally distributed.

Question 6: Testing a Hypothesised Population Distribution

A bank in Tanzania claims that 50% of its customers prefer mobile banking, 30% prefer ATM services and 20% prefer visiting bank branches. A sample of 200 customers gives the following results:

Banking method	Observed
Mobile banking	110
ATM	50
Bank branch	40
Total	**200**

Test the bank's claim at 5%.

Solution

Hypotheses:

- H_0: Customer preferences follow the stated proportions.
- H_1: Customer preferences do not follow the stated proportions.

Expected frequencies:

$$E_1 = 200(0.5) = 100$$

$$E_2 = 200(0.3) = 60$$

$$E_3 = 200(0.2) = 40$$

Method	O	E	Contribution
Mobile	110	100	1.000
ATM	50	60	1.667
Branch	40	40	0.000
Total	**200**	**200**	**2.667**

$$\chi^2 = 2.667$$

$$df = 3 - 1 = 2$$

Critical value = 5.991.

Since $2.667 < 5.991$, fail to reject H_0.

Conclusion: There is insufficient evidence to conclude that customer preferences differ from the bank's stated proportions.

Question 7: Distribution of Examination Grades

A college examines 200 students and classifies their performance into five categories.

Grade	Observed
A	50
B	35
C	45
D	30
F	40
Total	**200**

Test whether students are equally distributed across the five grades.

Solution

$$E = \frac{200}{5} = 40$$

Grade	O	E	Contribution
A	50	40	2.500
B	35	40	0.625
C	45	40	0.625
D	30	40	2.500
F	40	40	0.000
Total	**200**	**200**	**6.250**

$$\chi^2 = 6.25, \quad df = 5 - 1 = 4$$

Critical value = 9.488.

Since $6.25 < 9.488$, fail to reject H_0.

Conclusion: There is insufficient evidence that students are unequally distributed across the five grade categories.

Question 8: Customer Satisfaction Levels

A supermarket in Dar es Salaam surveys 200 customers about satisfaction levels.

Satisfaction	Observed
Very satisfied	30
Satisfied	35
Neutral	85
Dissatisfied	50
Total	**200**

The supermarket expects 10%, 20%, 40% and 30%, respectively. Test whether the observed distribution agrees with the expected distribution.

Solution

Expected frequencies:

$$E = (20, 40, 80, 60)$$

Category	O	E	Contribution
Very satisfied	30	20	5.000
Satisfied	35	40	0.625
Neutral	85	80	0.313

Category	O	E	Contribution
Dissatisfied	50	60	1.667
Total	**200**	**200**	**7.605**

$$\chi^2 = 7.605$$

$$df = 4 - 1 = 3$$

Critical value = 7.815.

Since $7.605 < 7.815$, fail to reject H_0.

Conclusion: At the 5% significance level, there is insufficient evidence that the observed satisfaction distribution differs from the supermarket's expected distribution.

3. Chi-Square Test of Independence

The Chi-square test of independence determines whether two categorical variables are statistically associated.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2(3) = 5.04, \quad p = 0.169$$

The sample does not provide enough evidence that the four proportions differ.

Observed	A	B	C	D
Observed	18	22	27	33
Expected	25	25	25	25
Contribution	1.96	0.36	0.16	2.56

Total contribution = 5.04 (below 5% cutoff)

Question 9: Gender and Choice of Financial Institution

A researcher surveys 160 customers to establish whether gender is associated with the choice of financial institution.

Gender	Bank A	Bank B	Bank C	Total
Male	40	30	10	80
Female	20	40	20	80
Total	**60**	**70**	**30**	**160**

Test the association at 5%.

Solution

Hypotheses:

- H_0: Gender and choice of financial institution are independent.
- H_1: Gender and choice of financial institution are associated.

Expected frequency:

$$E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{160}$$

Expected frequencies:

Gender	Bank A	Bank B	Bank C
Male	30	35	15
Female	30	35	15

Chi-square contributions:

$$\chi^2 = \frac{(40-30)^2}{30} + \frac{(30-35)^2}{35} + \frac{(10-15)^2}{15} + \frac{(20-30)^2}{30} + \frac{(40-35)^2}{35} + \frac{(20-15)^2}{15}$$

$$\chi^2 = 11.429$$

Degrees of freedom:

$$df = (2-1)(3-1) = 2$$

Critical value = 5.991.

Since $11.429 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between gender and choice of financial institution.

Question 10: Training and Employment Status

A study involving 100 graduates examines the relationship between professional training and employment status.

Training	Employed	Unemployed	Total
Trained	30	20	50
Not trained	10	40	50
Total	**40**	**60**	**100**

Determine whether training is associated with employment status.

Solution

Hypotheses:

- H_0: Training and employment status are independent.
- H_1: Training and employment status are associated.

Expected frequencies:

$$E_{11} = \frac{50(40)}{100} = 20$$

$$E_{12} = \frac{50(60)}{100} = 30$$

The second row also has expected frequencies of 20 and 30.

Training	Employed (O/E)	Unemployed (O/E)
Trained	30/20	20/30
Not trained	10/20	40/30

$$\chi^2 = \frac{(30-20)^2}{20} + \frac{(20-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(40-30)^2}{30}$$

$$\chi^2 = 16.667$$

$$df = (2-1)(2-1) = 1$$

Critical value = 3.841.

Since $16.667 > 3.841$, reject H_0.

Conclusion: Professional training is significantly associated with employment status in the surveyed graduates. This association does not, by itself, establish causation.

Question 11: Education Level and Loan Approval

A microfinance institution in Tanzania investigates whether education level is associated with loan approval.

Education	Approved	Rejected	Total
Primary	30	20	50
Secondary	40	10	50
University	20	30	50
Total	**90**	**60**	**150**

Test the relationship at 5%.

Solution

Expected frequencies for each row:

$$E_{\text{approved}} = \frac{50(90)}{150} = 30$$

$$E_{\text{rejected}} = \frac{50(60)}{150} = 20$$

Expected table:

Education	Approved	Rejected
Primary	30	20
Secondary	30	20
University	30	20

$$\chi^2 = 0 + 0 + \frac{(40-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(20-30)^2}{30} + \frac{(30-20)^2}{20}$$

$$\chi^2 = 16.667$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $16.667 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between education level and loan approval.

Question 12: Region and Preferred Payment Method

A researcher surveys 180 customers from three regions in Tanzania.

Region	Cash	Mobile Money	Bank Transfer	Total
Dar es Salaam	20	35	5	60
Dodoma	30	20	10	60
Mwanza	25	25	10	60
Total	**75**	**80**	**25**	**180**

Test whether region and payment method are independent.

Solution

Hypotheses:

- H_0: Region and payment method are independent.
- H_1: Region and payment method are associated.

Expected frequencies for each region:

$$E_{\text{cash}} = \frac{60(75)}{180} = 25$$

$$E_{\text{mobile}} = \frac{60(80)}{180} = 26.667$$

$$E_{\text{transfer}} = \frac{60(25)}{180} = 8.333$$

Expected table:

Region	Cash	Mobile	Transfer
Dar es Salaam	25	26.667	8.333
Dodoma	25	26.667	8.333
Mwanza	25	26.667	8.333

Substituting all nine cells:

$$\chi^2 \approx 12.525$$

$$df = (3-1)(3-1) = 4$$

Critical value = 9.488.

Since $12.525 > 9.488$, reject H_0.

Conclusion: Region is significantly associated with customers' preferred payment methods.

Question 13: Gender and Employment Preference

A survey of 120 graduates examines whether gender is associated with preference for public or private sector employment.

Gender	Public	Private	Total
Male	35	25	60
Female	20	40	60
Total	**55**	**65**	**120**

Solution

Expected frequencies:

$$E_{11} = \frac{60(55)}{120} = 27.5$$

$$E_{12} = \frac{60(65)}{120} = 32.5$$

Both rows have expected frequencies of 27.5 and 32.5.

$$\chi^2 = \frac{(35-27.5)^2}{27.5} + \frac{(25-32.5)^2}{32.5} + \frac{(20-27.5)^2}{27.5} + \frac{(40-32.5)^2}{32.5}$$

$$\chi^2 = 7.552$$

$$df = 1$$

Critical value = 3.841.

Since $7.552 > 3.841$, reject H_0.

Conclusion: There is a significant association between gender and employment preference.

Question 14: Advertising Method and Product Purchase

A business surveys 120 customers to examine whether advertising method is associated with purchasing behaviour.

Advertising	Purchased	Not purchased	Total
Television	35	15	50
Social media	40	10	50
Radio	10	10	20
Total	**85**	**35**	**120**

Solution

Expected frequencies:

Advertising	Purchased	Not purchased
Television	35.417	14.583
Social media	35.417	14.583
Radio	14.167	5.833

Using the chi-square formula:

$$\chi^2 \approx 7.89$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $7.89 > 5.991$, reject H_0.

Conclusion: Advertising method is significantly associated with customers' purchasing behaviour.

Question 15: Gender and Participation in Entrepreneurship

A college surveys 100 students to determine whether gender is associated with participation in entrepreneurship activities.

Gender	Participating	Not participating	Total
Male	30	20	50
Female	15	35	50
Total	**45**	**55**	**100**

Solution

Expected frequencies:

$$E_{\text{participating}} = \frac{50(45)}{100} = 22.5$$

$$E_{\text{not participating}} = \frac{50(55)}{100} = 27.5$$

$$\chi^2 = \frac{(30-22.5)^2}{22.5} + \frac{(20-27.5)^2}{27.5} + \frac{(15-22.5)^2}{22.5} + \frac{(35-27.5)^2}{27.5}$$

$$\chi^2 = 9.091$$

$$df = 1$$

Critical value = 3.841.

Since $9.091 > 3.841$, reject H_0.

Conclusion: There is a significant association between gender and participation in entrepreneurship activities.

4. Advanced Applications of Chi-Square Tests

Question 16: Fertiliser Type and Crop Performance

A researcher in Morogoro examines whether fertiliser type is associated with crop performance.

Fertiliser	High yield	Low yield	Total
Fertiliser A	40	10	50
Fertiliser B	25	25	50
Fertiliser C	15	35	50

Fertiliser	High yield	Low yield	Total
Total	**80**	**70**	**150**

Test the association at 5%.

Solution

Expected frequencies:

$$E_{\text{high}} = \frac{50(80)}{150} = 26.667$$

$$E_{\text{low}} = \frac{50(70)}{150} = 23.333$$

For each row, the expected values are 26.667 and 23.333.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$
$$\approx 22.768$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $22.768 > 5.991$, reject H_0.

Conclusion: There is a significant association between fertiliser type and crop performance.

Question 17: Hospital and Patient Recovery

A study examines the relationship between hospital location and patient recovery status.

Hospital	Recovered	Not recovered	Total
Hospital A	45	15	60
Hospital B	30	30	60
Hospital C	20	40	60
Total	**95**	**85**	**180**

Solution

Expected frequencies:

$$E_{\text{recovered}} = \frac{60(95)}{180} = 31.667$$

$$E_{\text{not recovered}} = \frac{60(85)}{180} = 28.333$$

$$\chi^2 \approx 26.675$$

$$df = 2$$

Critical value = 5.991.

Since $26.675 > 5.991$, reject H_0.

Conclusion: Hospital and patient recovery status are significantly associated in this sample. The result does not establish that hospital location causes differences in recovery.

Question 18: Business Ownership and Access to Credit

A survey of 150 entrepreneurs examines whether business ownership type is associated with access to credit.

Ownership	Access to credit	No access	Total
Sole proprietorship	35	15	50
Partnership	30	20	50
Company	45	5	50
Total	**110**	**40**	**150**

Solution

Expected frequencies:

$$E_{\text{access}} = \frac{50(110)}{150} = 36.667$$

$$E_{\text{no access}} = \frac{50(40)}{150} = 13.333$$

$$\chi^2 \approx 10.341$$

$$df = 2$$

Critical value = 5.991.

Since $10.341 > 5.991$, reject H_0.

Conclusion: Business ownership type is significantly associated with access to credit.

Question 19: Training Programme and Academic Performance

A college evaluates whether participation in a training programme is associated with examination performance.

Training	Pass	Fail	Total
Participated	45	5	50

Training	Pass	Fail	Total
Not participated	30	20	50
Total	**75**	**25**	**100**

Solution

Expected frequencies:

$$E_{\text{pass}} = \frac{50(75)}{100} = 37.5$$

$$E_{\text{fail}} = \frac{50(25)}{100} = 12.5$$

$$\chi^2 = \frac{(45-37.5)^2}{37.5} + \frac{(5-12.5)^2}{12.5} + \frac{(30-37.5)^2}{37.5} + \frac{(20-12.5)^2}{12.5}$$

$$\chi^2 = 12.00$$

$$df = 1$$

Critical value = 3.841.

Since $12.00 > 3.841$, reject H_0.

Conclusion: Participation in the training programme is significantly associated with examination performance.

Question 20: Age Group and Mobile Banking Adoption

A bank investigates whether age group is associated with mobile banking adoption.

Age group	Uses mobile banking	Does not use	Total
18–30	45	5	50
31–45	35	15	50
Above 45	20	30	50
Total	**100**	**50**	**150**

Solution

Expected frequencies for each age group:

$$E_{\text{uses}} = \frac{50(100)}{150} = 33.333$$

$$E_{\text{does not use}} = \frac{50(50)}{150} = 16.667$$

$$\chi^2 \approx 30.90$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $30.90 > 5.991$, reject H_0.

Conclusion: Age group is significantly associated with mobile banking adoption.

Question 21: Marketing Strategy and Sales Performance

A business examines whether marketing strategy is associated with sales performance.

Marketing strategy	High sales	Low sales	Total
Online marketing	40	10	50
Traditional marketing	25	25	50
Combined marketing	45	5	50
Total	**110**	**40**	**150**

Solution

Expected frequencies for each row:

$$E_{\text{high}} = 36.667, \quad E_{\text{low}} = 13.333$$

$$\chi^2 \approx 22.955$$

$$df = 2$$

Critical value = 5.991.

Since $22.955 > 5.991$, reject H_0.

Conclusion: Marketing strategy is significantly associated with sales performance.

Question 22: Type of Transport and Employee Punctuality

A company in Dar es Salaam studies the relationship between employees' transport methods and punctuality.

Transport	Punctual	Not punctual	Total
Bus	30	30	60
Private car	35	5	40
Motorcycle	25	15	40
Total	**90**	**50**	**140**

Solution

Expected frequencies:

Transport	Punctual	Not punctual
Bus	38.571	21.429
Private car	25.714	14.286
Motorcycle	25.714	14.286

Substituting into the chi-square formula:

$$\chi^2 \approx 16.79$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $16.79 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between transport method and employee punctuality.

Question 23: Agricultural Region and Crop Type

A study examines whether the type of crop cultivated is associated with agricultural region.

Region	Maize	Rice	Beans	Total
Morogoro	30	20	10	60
Mbeya	20	30	10	60
Dodoma	35	10	15	60
Total	**85**	**60**	**35**	**180**

Solution

Expected frequencies for each region:

$$E_{\text{maize}} = \frac{60(85)}{180} = 28.333$$

$$E_{\text{rice}} = \frac{60(60)}{180} = 20$$

$$E_{\text{beans}} = \frac{60(35)}{180} = 11.667$$

Expected table:

Region	Maize	Rice	Beans
Morogoro	28.333	20	11.667

Region	Maize	Rice	Beans
Mbeya	28.333	20	11.667
Dodoma	28.333	20	11.667

$$\chi^2 \approx 20.36$$

$$df = (3-1)(3-1) = 4$$

Critical value = 9.488.

Since $20.36 > 9.488$, reject H_0.

Conclusion: Agricultural region is significantly associated with crop type.

Question 24: Customer Loyalty and Service Quality

A telecommunications company surveys 120 customers to examine whether perceived service quality is associated with customer loyalty.

Service quality	Loyal	Not loyal	Total
High	40	10	50
Medium	25	15	40
Low	10	20	30
Total	**75**	**45**	**120**

Solution

Expected frequencies:

Service quality	Loyal	Not loyal
High	31.25	18.75
Medium	25	15
Low	18.75	11.25

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 \approx 23.64$$

$$df = (3-1)(2-1) = 2$$

Critical value = 5.991.

Since $23.64 > 5.991$, reject H_0.

Conclusion: There is a statistically significant association between perceived service quality and customer loyalty.

Question 25: Comprehensive Application – Education and Employment Status

A researcher investigates whether educational qualification is associated with employment status among 180 graduates in Tanzania.

The following data are collected:

Qualification	Employed	Unemployed	Self-employed	Total
Certificate	20	25	15	60
Diploma	30	15	15	60
Degree	40	10	10	60
Total	**90**	**50**	**40**	**180**

Required:

1. State the null and alternative hypotheses.
2. Calculate expected frequencies.
3. Calculate the chi-square statistic.
4. Determine the degrees of freedom.
5. Test the hypothesis at 5% significance.
6. Interpret the findings.

Solution

Step 1: Hypotheses

$$H_0 : \text{Educational qualification and employment status are independent.}$$

$$H_1 : \text{Educational qualification and employment status are associated.}$$

Step 2: Calculate expected frequencies

For certificate holders who are employed:

$$E_{11} = \frac{60(90)}{180} = 30$$

For certificate holders who are unemployed:

$$E_{12} = \frac{60(50)}{180} = 16.667$$

For certificate holders who are self-employed:

$$E_{13} = \frac{60(40)}{180} = 13.333$$

The same expected frequencies apply to the diploma and degree rows because each row has a total of 60.

Expected table:

Qualification	Employed	Unemployed	Self-employed
Certificate	30	16.667	13.333
Diploma	30	16.667	13.333
Degree	30	16.667	13.333

Step 3: Calculate chi-square

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

For certificate holders:

$$\frac{(20-30)^2}{30} + \frac{(25-16.667)^2}{}$$